



Risk-Averse Humanitarian Logistics

A CVaR Approach to Disaster Relief Networks

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Abstract

Standard disaster relief planning minimises expected cost or average unmet demand and can fail catastrophically in the extreme scenarios that dominate humanitarian impact. This paper extends classical humanitarian logistics models to a multi-period, multi-hub, multi-commodity network with two transport modes (truck and air) and introduces a mean-CVaR (mean-Conditional-Value-at-Risk) objective on total unmet demand that explicitly hedges against worst-case shortages. The model is instantiated on the Rawls and Turnquist (2010) Gulf Coast case study with fifteen historical hurricanes and solved with Gurobi. Moving from the risk-neutral baseline to a calibrated risk-averse policy raises expected cost by 72% but cuts $\text{CVaR}_{0.95}$ of unmet demand by 85% and lifts coverage in the worst Category-5 hurricane from 19.6% to 88.2%, with 76.5% of that coverage delivered in the first response period. The trade-off between expected efficiency and tail reliability is presented as a Pareto curve over the risk-aversion weight λ , giving planners a transparent decision aid.

1 Introduction

1.1 Background and Motivation

Disasters such as earthquakes, floods, and pandemics continue to pose significant humanitarian challenges worldwide. In the aftermath of a disaster, affected populations require immediate access to essential resources, including food, water, medicine, and emergency shelter. The ability of humanitarian organizations to deliver these resources quickly and effectively can significantly influence survival rates and the overall recovery process.

Humanitarian logistics plays a central role in disaster management by coordinating the procurement, storage, transportation, and distribution of relief supplies. Effective logistics operations are critical to reducing human suffering and ensuring that aid reaches affected populations in a timely manner under highly uncertain and resource-constrained conditions. Consequently, improving the planning and management of humanitarian logistics systems has become an important area of research within operations research and supply chain management.

1.2 Challenges in Humanitarian Logistics

Disaster relief operations are conducted in highly dynamic and uncertain environments, requiring relief organizations to make strategic and operational decisions with limited information. In the immediate aftermath of a disaster, decision-makers may lack accurate information regarding the location and severity of the event, the number of affected individuals, and the availability of critical transportation infrastructure.

These challenges are further exacerbated by significant disruptions to relief operations. Demand for emergency supplies can vary considerably across affected regions, transportation networks may become partially inaccessible due to damaged roads or bridges, and available resources are often limited relative to urgent needs. These uncertainties complicate resource allocation decisions and make it difficult to guarantee effective disaster response.

1.3 Risk-Neutral and Risk-Averse Planning in Humanitarian Logistics

Many humanitarian logistics models seek to optimize system performance by minimizing expected costs or maximizing average service levels across a set of possible disaster scenarios. Such approaches are commonly referred to as risk-neutral because they focus on average outcomes without explicitly considering the severity of extreme events [3, 1].

While risk-neutral planning may provide efficient solutions under normal conditions, it can perform poorly during severe disasters. A solution that appears optimal on average may still result in substantial

shortages, delayed response times, or inadequate service coverage when demand exceeds expectations or transportation networks experience significant disruptions. In humanitarian contexts, these failures can have serious consequences because unmet demand may directly affect affected populations.

To address this limitation, risk-averse planning approaches have been proposed. Rather than focusing solely on expected performance, risk-averse models explicitly account for the potential impact of worst-case scenarios. As a result, these approaches seek to improve the system reliability and reduce vulnerability to catastrophic events.

1.4 Research Objective

Motivated by the challenges of disaster response under uncertainty, this seminar paper investigates the application of risk-aware optimization in humanitarian logistics. In particular, the study examines how incorporating Conditional Value-at-Risk (CVaR) into disaster relief planning influences logistics performance when compared with traditional risk-neutral approaches.

The primary research question addressed in this paper is:

How does shifting from a risk-neutral optimization approach to a CVaR-based risk-averse optimization approach affect unmet demand and response times in humanitarian disaster relief planning?

2 Literature Review

Humanitarian logistics under uncertainty has been studied through stochastic programming models that differ mainly in how they represent time, network structure, and attitude toward risk. This section reviews the three anchor papers approved for our seminar topic and relates them to the risk-neutral versus risk-averse comparison developed in later sections.

2.1 Pre-Positioning under Disaster Uncertainty

Rawls and Turnquist [1] formulate a two-stage stochastic mixed-integer program for pre-positioning emergency supplies before hurricanes or similar disasters. First-stage decisions select facility locations and stock levels. Second-stage decisions allocate supplies after a scenario is realised. Uncertainty enters through scenario-dependent demand and transportation network availability. The objective minimises *expected* total cost, which makes the model risk-neutral. It performs well on average but does not penalise solutions that fail severely in low-probability, high-impact scenarios. Rawls and Turnquist illustrate the approach on a Gulf Coast case study and develop a Lagrangian L-shaped heuristic for large instances. Our work adopts their pre-positioning logic (facility opening, inventory positioning, and scenario-dependent flows) as the structural backbone of the network model.

2.2 Multi-Period Disaster Logistics

Yi and Özdamar [2] propose an integrated location and distribution model for coordinating relief shipments and evacuation support over multiple periods. Their formulation uses a multi-commodity network flow structure in which vehicles are treated as flow commodities, enabling dynamic re-planning as information updates arrive after a disaster. Although their application emphasises patient transport and medical staffing, the multi-period perspective is directly relevant to sustained relief operations in which demand evolves over days. We adopt a discrete planning horizon and period-indexed flows. Our implementation adds truck and air modes with scenario-dependent capacities rather than explicit vehicle routing.

2.3 Risk-Averse Stochastic Programming and CVaR

Noyan [3] extends classical two-stage stochastic programming by replacing the expectation-only objective with a mean-CVaR criterion. Conditional Value-at-Risk (CVaR) measures the expected loss in the worst $(1 - \alpha)$ tail of the loss distribution and can be linearised via auxiliary variables, following the formulation of Rockafellar and Uryasev. Noyan applies this risk-averse model to disaster preparedness, jointly selecting response facility locations and inventory levels under uncertain demand and network damage. Her results show that incorporating CVaR shifts pre-positioning decisions toward configurations that reduce exposure to catastrophic outcomes, at some increase in expected cost. This paper provides the methodological foundation for our CVaR extension in Section 4.9.

2.4 Risk-Neutral versus Risk-Averse Planning

Table 1 summarises how the three anchor papers relate to our contribution. Rawls and Turnquist supply the humanitarian network context. Yi and Özdamar supply the multi-period planning horizon. Noyan supplies the risk-averse objective. Together they motivate a model that is risk-neutral when $\lambda = 0$ and risk-averse when a weighted CVaR term is added with $\lambda > 0$.

Following our approved seminar scope and confirmation from the course instructor, we focus on mean-CVaR stochastic programming. Distributionally robust formulations (for example Lin et al. 2018, mentioned in the original topic description) are outside our scope because they require ambiguity sets and different algorithmic machinery. The original topic sheet also misstates some bibliographic details. The papers listed above are the correct, instructor-approved references.

Table 1: Comparison of anchor papers and our extension

Reference	Main contribution	Role in our paper
Rawls & Turnquist (2010)	Two-stage SMIP for supply pre-positioning under demand and network uncertainty	Network structure, first/second-stage decisions, scenario data
Yi & Özdamar (2007)	Multi-period integrated location and distribution for disaster response	Multi-period flows and dynamic demand horizon
Noyan (2012)	Mean-CVaR two-stage stochastic programming for disaster preparedness	CVaR objective, risk-neutral vs risk-averse comparison
This paper	Multi-period hub network with modes, travel times, mean-CVaR on shortages	Combines the three elements and adds computational experiments

The key trade-off identified across this literature is between *expected efficiency* and *tail reliability*. Risk-neutral models minimise average cost but may accept high unmet demand in extreme scenarios. Risk-averse models sacrifice some expected performance to limit worst-case exposure. Section 6 quantifies this trade-off on the Rawls Gulf Coast case study.

3 Risk Measures for Humanitarian Logistics

This section introduces the two risk measures used throughout the paper, Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR), and explains why we build the risk-averse policy around CVaR of *unmet demand* rather than CVaR of cost. The section is deliberately written at a level accessible to readers without a background in stochastic programming.

3.1 Random Loss and Its Distribution

Let L denote a random *loss* associated with a policy: for a humanitarian planner, L can be a monetary cost, a number of undelivered relief units, or any other outcome the planner wants to keep small. Once a policy is chosen, uncertainty about the disaster (severity, damage, demand) makes L a random variable with a probability distribution $\mathbb{P}(L \leq \ell)$. Traditional risk-neutral planning collapses this distribution into a single number, its *expectation* $\mathbb{E}[L]$, and picks the policy that minimises it. Two policies with the same expected loss can therefore look identical to a risk-neutral planner even when one of them has a much heavier right tail.

3.2 Value-at-Risk (VaR)

The Value-at-Risk of L at confidence level $\alpha \in (0, 1)$ (we use $\alpha = 0.95$) is the smallest threshold η such that the probability of L exceeding η is at most $1 - \alpha$:

$$\text{VaR}_\alpha(L) = \inf\{\eta \in \mathbb{R} : \mathbb{P}(L > \eta) \leq 1 - \alpha\}. \quad (1)$$

Intuitively, in 95% of scenarios the loss stays below $\text{VaR}_{0.95}(L)$, and in the remaining 5% it may exceed this value by an unspecified amount. VaR is widely used in finance but has two drawbacks that matter here. First, it is silent about *how bad* the worst 5% of outcomes can be. Second, it is not a coherent risk measure. It can violate sub-additivity, meaning diversification can appear to *increase* VaR. For a humanitarian planner, ignoring how bad the tail can get is exactly the wrong thing to do.

3.3 Conditional Value-at-Risk (CVaR)

Conditional Value-at-Risk fixes both drawbacks by averaging the loss over the worst $(1 - \alpha)$ portion of the distribution:

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]. \quad (2)$$

CVaR is coherent (in particular, sub-additive), always at least as large as VaR at the same α , and directly answers the question “what is the expected outcome in the bad tail?”. In this paper we use $\alpha = 0.95$, so $\text{CVaR}_{0.95}$ of unmet demand is the expected number of relief units that fail to reach demand nodes in the worst 5% of hurricanes weighted by their probability.

3.4 Linear Representation for Optimisation

To use CVaR inside a mixed-integer programme we need a linear representation. Rockafellar and Uryasev [4] prove that CVaR equals the optimal value of a convex programme in an auxiliary threshold η :

$$\text{CVaR}_\alpha(L) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E}[(L - \eta)^+] \right\}, \quad (3)$$

where $(x)^+ = \max(x, 0)$. The argument is short. For a fixed η , the expression $\eta + \frac{1}{1 - \alpha} \mathbb{E}[(L - \eta)^+]$ is always at least as large as $\text{CVaR}_\alpha(L)$, and equality is attained when η is chosen as a $\text{VaR}_\alpha(L)$ threshold. Minimising over η therefore recovers CVaR exactly. The positive-part operator is not linear, but it admits an epigraph representation. Introduce a non-negative slack ξ for each realisation of L and require $\xi \geq L - \eta$. Then ξ can be driven down to $(L - \eta)^+$ at optimality, so

$$\text{CVaR}_\alpha(L) = \min_{\eta, \xi} \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E}[\xi] : \xi \geq L - \eta, \xi \geq 0 \right\}. \quad (4)$$

In the discrete scenario setting used here, with scenarios $s \in S$, probabilities p_s , and scenario losses L_s , the expectation becomes a finite sum and each scenario gets its own slack ξ_s :

$$\text{CVaR}_\alpha(L) = \min_{\eta, \xi_s} \left\{ \eta + \frac{1}{1 - \alpha} \sum_{s \in S} p_s \xi_s : \xi_s \geq L_s - \eta, \xi_s \geq 0 \right\}. \quad (5)$$

Equation (5) is linear in the decision variables η and ξ_s . Substituting the scenario shortage $L_s = S_s$ into this form and adding the expected-cost term $\sum_s p_s Z_s$ with weight λ yields the mean-CVaR objective of Section 4.9, which is exactly the objective solved by Gurobi.

3.5 Loss on Cost vs. Loss on Unmet Demand

The remaining choice is what to make L . The natural stochastic-programming default is monetary cost, so that $L_s = Z_s$ and the objective penalises expensive tail scenarios. In a humanitarian setting this default has two problems. First, it can be strongly influenced by a single cost component. For example, a large penalty on unmet demand can push CVaR of cost around without visibly moving service. Second, it does not directly answer the question a relief manager cares about, namely how much aid is left undelivered in the worst hurricanes. We therefore set $L_s = S_s$, the total unmet demand in scenario s , so that $\text{CVaR}_\alpha(S)$ is the expected shortage in the worst $(1 - \alpha)$ share of scenarios. Cost still enters the objective through the expected-cost term $\mathbb{E}[Z]$, and the weight λ controls the trade-off between them. Section 4.9 formalises this choice.

3.6 Risk-Neutral vs. Risk-Averse Planning

A risk-neutral policy is recovered as the special case $\lambda = 0$, in which the CVaR term drops out and the model minimises expected cost alone. A risk-averse policy sets $\lambda > 0$, trading a rise in expected cost for a fall in CVaR of unmet demand. Larger λ places more weight on the tail. Because the model is a mixed-integer linear programme in every λ , we can sweep λ to trace an efficiency and resilience Pareto curve, as done in Section 6.

4 Mathematical Formulation

4.1 Problem Setting

We formulate a multi-period stochastic humanitarian logistics model for disaster relief planning. The model determines pre-disaster facility location and inventory pre-positioning decisions, as well as post-disaster commodity distribution decisions. Uncertainty is represented by a finite set of disaster scenarios, which capture uncertain demand, damage to pre-positioned supplies, and disruption of

transportation capacity. The model is first written as a risk-neutral stochastic program that minimizes expected total cost. Then, a Conditional Value-at-Risk (CVaR) term is added to obtain a risk-averse formulation that protects against high-cost disaster scenarios.

The model follows a two-stage decision structure. First-stage decisions are made before the disaster scenario is known and include facility opening and inventory pre-positioning decisions. Second-stage decisions are made after the scenario is realized and include shipment quantities, remaining inventory, and unmet demand over the planning horizon.

This formulation builds on the emergency supply pre-positioning framework of Rawls and Turnquist [1], the dynamic disaster logistics model of Yi and Özdamar [2], and the risk-averse CVaR stochastic programming approach of Noyan [3].

4.2 Model Scope and Assumptions

The model focuses on relief-supply pre-positioning and distribution at candidate *hub* locations. Two transportation modes are available on each arc: road convoys (truck) and emergency airlifts (air). Road shipments require one period to arrive, while air shipments arrive within the same period. We do not model individual vehicle routes or patient evacuation.

The model is based on the following assumptions:

1. Disaster uncertainty is represented by a finite set of scenarios $s \in S$, each with probability p_s .
2. Facility location and inventory pre-positioning decisions are made before the disaster scenario is known.
3. Distribution, inventory, and shortage decisions are scenario-dependent and are made after the disaster scenario is realized.
4. Multiple relief commodities may be pre-positioned and transported.
5. Demand that cannot be satisfied is allowed, but it is penalized in the objective function.
6. Transportation capacity and mode availability may vary by scenario because of infrastructure damage.
7. A fraction of pre-positioned inventory may be damaged or unavailable after the disaster. Secondary hubs may be more severely affected in extreme scenarios.
8. Truck shipments dispatched in period t arrive at the start of period $t+1$. Air shipments dispatched in period t arrive in the same period.

9. Dynamic inventory tracking allows items to be carried over between periods, but physical inventory storage is restricted to opened hub locations.
10. The risk-averse model adds a weighted CVaR term on total unmet demand to limit tail shortages.

4.3 Sets and Indices

Table 2: Sets and indices

Symbol	Description
N	Set of all nodes in the relief network.
$I \subseteq N$	Set of candidate facility locations.
$D \subseteq N$	Set of demand locations.
A	Set of directed transportation arcs (i, j) .
K	Set of relief commodities.
$T = \{1, \dots, H\}$	Set of operational time periods in the response horizon.
$T_0 = \{0\} \cup T$	Set of inventory time periods, including the initial pre-disaster period.
S	Set of disaster scenarios.
L	Set of facility size options.
M	Set of transportation modes (truck, air).

4.4 Parameters

Table 3: Model parameters

Symbol	Description
p_s	Probability of scenario s , where $\sum_{s \in S} p_s = 1$.
d_{ikts}	Demand for commodity k at node i in period t under scenario s . We set $d_{ikts} = 0$ for all $i \notin D$.
γ_{iks}	Fraction of commodity k pre-positioned at node i that remains usable under scenario s , where $0 \leq \gamma_{iks} \leq 1$.
U_{ijmts}	Transportation capacity on arc (i, j) by mode m in period t under scenario s .
F_{il}	Fixed cost of opening a facility of size l at hub i .
M_l	Storage capacity of a facility of size l .
b_k	Storage space required per unit of commodity k .
q_k	Acquisition cost per unit of commodity k .
c_{ijkmts}	Unit transportation cost for commodity k on arc (i, j) by mode m in period t under scenario s .
e_k	Capacity consumption per unit of commodity k on transportation arcs.
h_k	Holding cost per unit of remaining inventory of commodity k .
ρ_k	Penalty cost per unit of unmet demand of commodity k .
τ_m	Delivery lead time (in periods) for mode $m \in M$.
α	CVaR confidence level, where $0 < \alpha < 1$.
λ	Risk-aversion weight, where $\lambda \geq 0$.

4.5 Decision Variables

The model contains first-stage, second-stage, and CVaR auxiliary variables. Their meanings are summarized in Table 4. Domain restrictions are explicitly stated within the full MILP formulation equations.

Table 4: Decision variables

Variable	Type	Description
y_{il}	First-stage	Binary variable equal to 1 if a facility of size l is opened at node i , and 0 otherwise.
r_{ik}	First-stage	Amount of commodity k pre-positioned at candidate facility node i .
x_{ijkmts}	Second-stage	Amount of commodity k shipped from node i to node j using mode m in period t under scenario s .
I_{ikts}	Second-stage	Inventory of commodity k at node i at the end of period t under scenario s .
w_{ikts}	Second-stage	Unmet demand of commodity k at node i in period t under scenario s .
η	CVaR	Value-at-Risk threshold variable for total unmet demand across scenarios.
ξ_s	CVaR	Excess unmet demand above the Value-at-Risk threshold in scenario s .
Z_s	Auxiliary	Total realized first- and second-stage cost under scenario s .
S_s	Auxiliary	Total unmet demand summed across demand nodes, commodities, and periods under scenario s .

4.6 Cost Definitions

The first-stage cost consists of facility opening costs and commodity acquisition costs:

$$C^{\text{first}} = \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{i \in I} \sum_{k \in K} q_k r_{ik}. \quad (6)$$

For each scenario s , the second-stage operational cost includes transportation cost across all modes $m \in M$, inventory holding cost, and a penalty on unmet demand at demand nodes:

$$Q_s = \sum_{t \in T} \sum_{(i,j) \in A} \sum_{k \in K} \sum_{m \in M} c_{ijkmts} x_{ijkmts} + \sum_{t \in T} \sum_{i \in I} \sum_{k \in K} h_k I_{ikts} + \sum_{t \in T} \sum_{i \in D} \sum_{k \in K} \rho_k w_{ikts}, \quad \forall s \in S. \quad (7)$$

The total realized cost under scenario s is the summation of both stages:

$$Z_s = C^{\text{first}} + Q_s, \quad \forall s \in S. \quad (8)$$

4.7 Risk-Neutral Stochastic Model

The baseline risk-neutral model minimizes expected total cost across all scenarios:

$$\min \quad C^{\text{first}} + \sum_{s \in S} p_s Q_s. \quad (9)$$

Equivalently, this can be written as:

$$\min \quad \sum_{s \in S} p_s Z_s. \quad (10)$$

4.8 Model Constraints

4.8.1 Facility Capacity and Storage Bounds

The amount of inventory pre-positioned at a candidate facility cannot exceed the capacity of the selected facility size:

$$\sum_{k \in K} b_k r_{ik} \leq \sum_{l \in L} M_l y_{il}, \quad \forall i \in I. \quad (11)$$

Inventory held dynamically at an opened facility cannot exceed its selected capacity:

$$\sum_{k \in K} b_k I_{ikts} \leq \sum_{l \in L} M_l y_{il}, \quad \forall i \in I, t \in T, s \in S. \quad (12)$$

For nodes that are not designated facility locations, physical inventory storage is strictly banned:

$$I_{ikts} = 0, \quad \forall i \in N \setminus I, k \in K, t \in T_0, s \in S. \quad (13)$$

4.8.2 Facility Selection

At most one facility size can be selected at each candidate location:

$$\sum_{l \in L} y_{il} \leq 1, \quad \forall i \in I. \quad (14)$$

4.8.3 Initial Usable Inventory

Only the fraction of pre-positioned inventory that survives the disaster is available at the beginning of the response phase. Period $t = 0$ denotes the pre-disaster initialization state, which sits outside the operational planning horizon T :

$$I_{ik0s} = \gamma_{iks} r_{ik}, \quad \forall i \in I, k \in K, s \in S. \quad (15)$$

4.8.4 Inventory Balance with Multi-Mode Flows and Delivery Delays

Shipments are indexed by transportation mode $m \in M$. Truck shipments dispatched in period t arrive at the start of period $t + \tau_{\text{truck}}$. Air shipments arrive in the same period, so $\tau_{\text{air}} = 0$. Material flows are balanced at every node using a single equation that folds in both modes:

$$I_{ikts} = I_{ik,t-1,s} + \sum_{j:(j,i) \in A} \sum_{m \in M} x_{jikm,t-\tau_m,s} - \sum_{j:(i,j) \in A} \sum_{m \in M} x_{ijkmts} - d_{ikts} + w_{ikts}, \quad (16)$$

$$\forall i \in N, k \in K, t \in T, s \in S.$$

Shipments with a departure period $t - \tau_m$ outside the planning horizon T are treated as zero, so period-1 inflows can only come from air.

4.8.5 Shortage Bounds

Unmet demand cannot exceed actual demand at demand nodes:

$$0 \leq w_{ikts} \leq d_{ikts}, \quad \forall i \in D, k \in K, t \in T, s \in S. \quad (17)$$

For locations outside the set of demand destinations, unmet demand variables are identically bounded to zero:

$$w_{ikts} = 0, \quad \forall i \in N \setminus D, k \in K, t \in T, s \in S. \quad (18)$$

4.8.6 Transportation Capacity

The total load shipped on each arc using a given mode cannot exceed the scenario-dependent mode-specific capacity:

$$\sum_{k \in K} e_k x_{ijkmts} \leq U_{ijmts}, \quad \forall (i,j) \in A, m \in M, t \in T, s \in S. \quad (19)$$

The case study in Section 5 sets $U_{ijmts} = 0$ whenever hurricane s knocks out the road link (i,j) . Air capacity on the same arc is left unaffected because emergency airlifts do not depend on the road network.

4.9 CVaR Risk-Averse Extension

Following Noyan [3] and the general linearisation of Rockafellar and Uryasev introduced in Section 3, we keep the expected cost $\mathbb{E}[Z]$ as the primary efficiency term and add a Conditional Value-at-Risk

penalty on total unmet demand. Define the scenario shortage:

$$S_s = \sum_{t \in T} \sum_{i \in D} \sum_{k \in K} w_{ikts}, \quad \forall s \in S. \quad (20)$$

The mean-CVaR objective used in the computational experiments is:

$$\min \sum_{s \in S} p_s Z_s + \lambda \left[\eta + \frac{1}{1 - \alpha} \sum_{s \in S} p_s \xi_s \right]. \quad (21)$$

with

$$\xi_s \geq S_s - \eta, \quad \forall s \in S, \quad (22)$$

$$\xi_s \geq 0, \quad \forall s \in S, \quad (23)$$

$$\eta \in \mathbb{R}. \quad (24)$$

When $\lambda = 0$, the model reduces to risk-neutral planning. Larger values of λ increase the weight on tail shortages relative to expected cost.

4.10 Complete Risk-Averse MILP Formulation

Collecting all foundational equations, the complete risk-averse mixed-integer linear programming model is stated as:

$$\min \sum_{s \in S} p_s Z_s + \lambda \left[\eta + \frac{1}{1 - \alpha} \sum_{s \in S} p_s \xi_s \right] \quad (25a)$$

$$\text{s.t. } Z_s = C^{\text{first}} + Q_s, \quad \forall s \in S, \quad (25b)$$

$$C^{\text{first}} = \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{i \in I} \sum_{k \in K} q_k r_{ik}, \quad (25c)$$

$$Q_s = \sum_{t \in T} \sum_{(i,j) \in A} \sum_{k \in K} \sum_{m \in M} c_{ijkmts} x_{ijkmts} + \sum_{t \in T} \sum_{i \in I} \sum_{k \in K} h_k I_{ikts} + \sum_{t \in T} \sum_{i \in D} \sum_{k \in K} \rho_k w_{ikts}, \quad \forall s \in S, \quad (25d)$$

$$S_s = \sum_{t \in T} \sum_{i \in D} \sum_{k \in K} w_{ikts}, \quad \forall s \in S, \quad (25e)$$

$$\sum_{k \in K} b_k r_{ik} \leq \sum_{l \in L} M_l y_{il}, \quad \forall i \in I, \quad (25f)$$

$$\sum_{k \in K} b_k I_{ikts} \leq \sum_{l \in L} M_l y_{il}, \quad \forall i \in I, t \in T, s \in S, \quad (25g)$$

$$I_{ikts} = 0, \quad \forall i \in N \setminus I, k \in K, t \in T_0, s \in S, \quad (25h)$$

$$\sum_{l \in L} y_{il} \leq 1, \quad \forall i \in I, \quad (25i)$$

$$I_{ik0s} = \gamma_{iks} r_{ik}, \quad \forall i \in I, k \in K, s \in S, \quad (25j)$$

$$I_{ikts} = I_{ik,t-1,s} + \sum_{j:(j,i) \in A} \sum_{m \in M} x_{jikm,t-\tau_m,s} - \sum_{j:(i,j) \in A} \sum_{m \in M} x_{ijkmts} - d_{ikts} + w_{ikts}, \\ \forall i \in N, k \in K, t \in T, s \in S, \quad (25k)$$

$$0 \leq w_{ikts} \leq d_{ikts}, \quad \forall i \in D, k \in K, t \in T, s \in S, \quad (25l)$$

$$w_{ikts} = 0, \quad \forall i \in N \setminus D, k \in K, t \in T, s \in S, \quad (25m)$$

$$\sum_{k \in K} e_k x_{ijkmts} \leq U_{ijmts}, \quad \forall (i,j) \in A, m \in M, t \in T, s \in S, \quad (25n)$$

$$\xi_s \geq S_s - \eta, \quad \forall s \in S, \quad (25o)$$

$$\xi_s \geq 0, \quad \forall s \in S, \quad (25p)$$

$$y_{il} \in \{0, 1\}, \quad \forall i \in I, l \in L, \quad (25q)$$

$$r_{ik} \geq 0, \quad \forall i \in I, k \in K, \quad (25r)$$

$$x_{ijkmts} \geq 0, \quad \forall (i,j) \in A, k \in K, m \in M, t \in T, s \in S, \quad (25s)$$

$$I_{ikts} \geq 0, \quad \forall i \in N, k \in K, t \in T_0, s \in S, \quad (25t)$$

$$w_{ikts} \geq 0, \quad \forall i \in N, k \in K, t \in T, s \in S, \quad (25u)$$

$$\eta \in \mathbb{R}. \quad (25v)$$

4.11 Performance Measures

To compare the risk-neutral and risk-averse models, we evaluate the following performance measures.

The expected total cost is:

$$\mathbb{E}[Z] = \sum_{s \in S} p_s Z_s. \quad (26)$$

The Conditional Value-at-Risk of total unmet demand, computed from the CVaR auxiliary variables, is:

$$\text{CVaR}_\alpha(S) = \eta + \frac{1}{1 - \alpha} \sum_{s \in S} p_s \xi_s. \quad (27)$$

This is the primary tail-risk indicator reported in Section 6. Lower values mean fewer units of demand are expected to go unmet in the worst $(1 - \alpha)$ share of scenarios.

The service coverage under scenario s across all commodities is defined as:

$$\text{Coverage}_s = 1 - \frac{\sum_{t \in T} \sum_{i \in D} \sum_{k \in K} w_{ikts}}{\sum_{t \in T} \sum_{i \in D} \sum_{k \in K} d_{ikts}}, \quad \forall s \in S. \quad (28)$$

The expected service coverage is:

$$\mathbb{E}[\text{Coverage}] = \sum_{s \in S} p_s \text{Coverage}_s. \quad (29)$$

To proxy response time, we report period-specific coverage in scenario s :

$$\text{Coverage}_{ts} = 1 - \frac{\sum_{i \in D} \sum_{k \in K} w_{ikts}}{\sum_{i \in D} \sum_{k \in K} d_{ikts}}, \quad \forall t \in T, s \in S. \quad (30)$$

Higher period-1 coverage in a severe scenario indicates faster initial response.

5 Case Study: Rawls Gulf Coast

To evaluate the risk-neutral and risk-averse policies on a benchmark rooted in real data rather than a hand-tuned synthetic instance, we build the computational study on the Gulf Coast test problem of Rawls and Turnquist [1]. Rawls and Turnquist provide, in their Tables 3 and 4, fifteen historical hurricanes with categories, landfall nodes, unusable road links, and per-commodity demand, together with the probability of each hurricane occurring. We restrict attention to the fifteen single-hurricane scenarios (Table 4 top block) and renormalise their probabilities by dividing by the published total of 0.75, so that our scenario probabilities sum to one. The remaining probability mass in the source paper covers combined-hurricane events that are outside the scope of this seminar.

5.1 Network, Hubs, and Commodities

The instance places thirty nodes along a Gulf-Coast-inspired coastal-to-inland layout. Eleven of these are candidate hub locations, chosen to mix vulnerable coastal sites with safer inland ones so that the model faces a real trade-off between proximity to demand and exposure to hurricane damage. The arc set is built by connecting node pairs whose Euclidean distance is below a fixed threshold, and every link that Rawls lists as unusable in at least one hurricane is force-included so that the effect of a road cut is always well defined. Three relief commodities are pre-positioned and shipped: water (thousands of gallons), food (thousands of units), and medicine (units). Their acquisition, holding, storage-space, and shortage-penalty parameters reflect the fact that water is bulky and cheap while medicine is compact and expensive per unit.

5.2 Hurricanes and Damage

Table 5 lists the fifteen Rawls hurricanes with their category, landfall nodes, unusable links, and total per-commodity demand. Following Rawls, hurricanes of category three or higher destroy all supplies stocked at a landfall node ($\gamma_{iks} = 0$), while category one and category two hurricanes leave half of the supplies usable ($\gamma_{iks} = 0.5$). Nodes that are not landfall points are unaffected. Each hurricane's aggregate demand is placed on its landfall nodes and split across the three operational periods with weights (0.50, 0.30, 0.20), reflecting an intense immediate phase followed by smaller short-term and long-term needs [2]. For the three hurricanes without a specific coastal landfall (10, 12, and 13 in Rawls' numbering), demand is spread evenly across all nodes because the storm affects a broad region without a single strike point. Unusable road links from Rawls zero the truck capacity on the affected arcs. Air transport is left available on the same arcs because emergency airlifts do not depend on the road network.

Table 5: Rawls single-hurricane scenarios used in the case study. Category, landfall, unusable links, and aggregate demand are taken verbatim from Rawls and Turnquist [1] Table 3. The probability column shows the raw Table 4 value and the renormalised value used here.

Scen.	Cat.	Landfall	Unusable links	Water	Food	Medicine	p_s raw	p_s norm.
h1	3	3	(4,5)	350	525	500	0.0231	0.0308
h2	5	14	(12,14),(14,15),(15,24)	560	927	883	0.0092	0.0123
h3	2	22	none	861	181	402	0.0536	0.0715
h4	2	22	(17,20)	9000	1692	3760	0.0536	0.0715
h5	4	11, 29	none	7500	1771	1687	0.0500	0.0667
h6	3	15	none	1000	1838	1751	0.0308	0.0411
h7	2	21	(21,22)	600	324	720	0.0536	0.0715
h8	1	11	(8,12)	1500	162	360	0.0308	0.0411
h9	5	13, 29	(12,13)	1040	13300	95000	0.0500	0.0667
h10	2	none	none	2250	1125	18750	0.1617	0.2156
h11	3	21	(21,22),(15,24)	5000	1750	12500	0.0230	0.0306
h12	3	none	none	18000	630	4500	0.0308	0.0411
h13	3	none	none	2818	80	571	0.1338	0.1784
h14	4	14, 30	none	2239	1477	10551	0.0230	0.0306
h15	4	22	none	4400	3921	28007	0.0230	0.0306

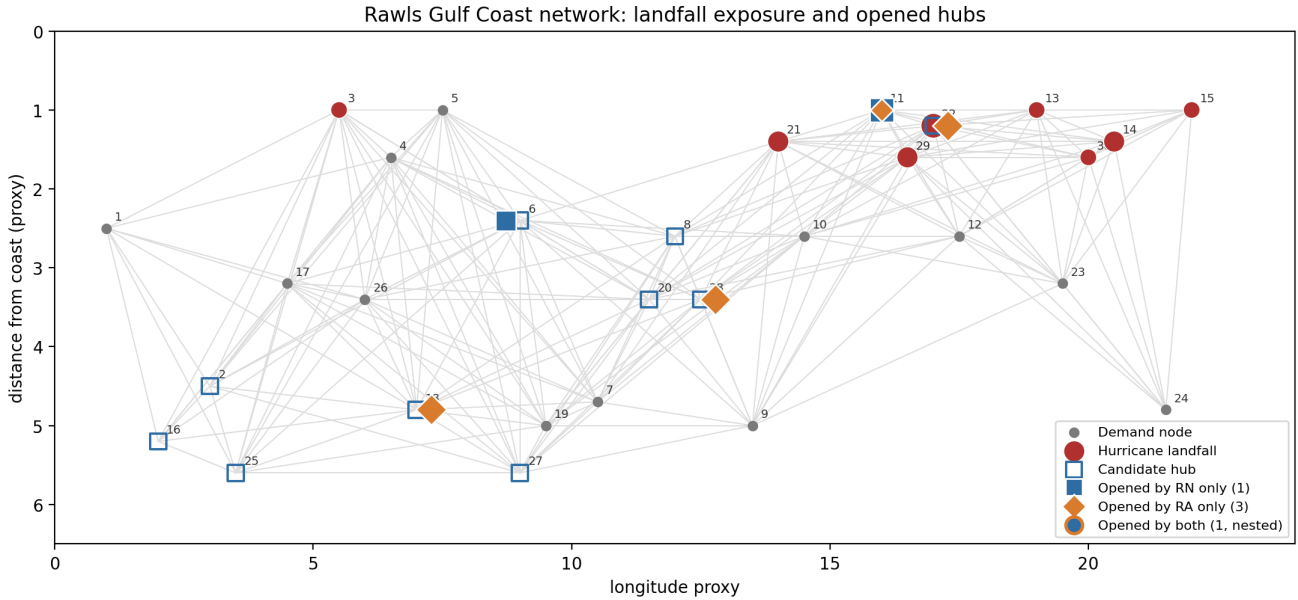


Figure 1: Rawls Gulf Coast network. Red circles mark landfall nodes and are sized by how many hurricanes strike that node. Blue outline squares are candidate hubs. Hubs opened by the risk-neutral policy only are shown as filled blue squares offset left. Hubs opened by the risk-averse policy only are shown as orange diamonds offset right. Hubs opened by both policies are drawn as a nested blue square with an orange diamond on top, so neither marker is hidden.

6 Computational Experiments

The extended MILP is implemented in Python with Gurobi 13. First-stage decisions (hub opening and pre-positioning) are scenario-independent. Flows, inventories, and shortages are scenario-dependent. The risk-neutral baseline sets $\lambda = 0$. The risk-averse headline policy uses the calibrated weight $\lambda = 18$ with $\alpha = 0.95$. All numerical values reported below are read from the CSV files written by `run_experiments.py` in the `results/` directory. No number in this section is hand-typed.

6.1 Implementation and Data

The Rawls instance is built by `src/data/rawls_instance.py` and saved as JSON for reproducibility. Figure 2 summarises the computational pipeline from the Rawls tables to the CSV metrics and figures used below. On this instance the MILP has roughly 220,000 continuous flow variables and 33 binary facility variables. Table 6 reports Gurobi wall-clock solve times on a laptop with a Named-User Academic licence. The risk-neutral solve finishes in 10.8 seconds. The headline risk-averse solve finishes in 13.4 seconds. Across the λ -sweep every solve stays between 11 and 14 seconds, so the full experiment set remains laptop-scale.

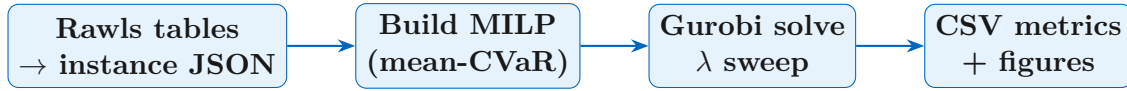


Figure 2: Computational pipeline. Every number in Section 6 is read from the CSV files produced by this pipeline.

Table 6: Gurobi solve times (seconds) for the risk-neutral policy, the headline risk-averse policy, and the λ -sweep. Values are taken from `results/experiment_summary.csv` and `results/lambda_sweep.csv`.

Policy / λ	Solve time (s)	$\mathbb{E}[Z]$	$\text{CVaR}_{0.95}(S)$	Hubs
Risk-neutral ($\lambda = 0$)	10.8	1,443,496	87,858	2
Risk-averse ($\lambda = 18$)	13.4	2,489,297	12,855	4
$\lambda = 0$	11.1	1,443,496	87,858	2
$\lambda = 5$	11.0	1,447,014	86,933	2
$\lambda = 10$	11.1	1,538,330	76,250	3
$\lambda = 15$	11.5	1,640,377	66,993	3
$\lambda = 18$	13.4	2,489,297	12,855	4
$\lambda = 22$	13.4	2,743,204	0	5

6.2 Policy Comparison

Table 7 and Figure 3 compare the two policies on the Rawls instance. Figure 4 shows coverage across all fifteen hurricanes, Figure 5 breaks the coverage in the worst hurricane down by commodity, Figure 6 plots period-specific coverage in the worst hurricane, and Figure 7 shows the empirical CDF of unmet demand across scenarios.

Table 7: Risk-neutral ($\lambda = 0$) vs. risk-averse ($\lambda = 18$) policies on the Rawls Gulf Coast instance.

Metric	Risk-neutral	Risk-averse
Expected total cost $\mathbb{E}[Z]$	1,443,496	2,489,297
$\text{CVaR}_{0.95}$ of unmet demand	87,858	12,855
Expected coverage	83.5%	93.4%
Coverage in worst hurricane (h9, Cat. 5)	19.6%	88.2%
Total unmet demand in h9	87,858	12,855
Period-1 coverage in h9	0.9%	76.5%
Number of hubs opened	2	4

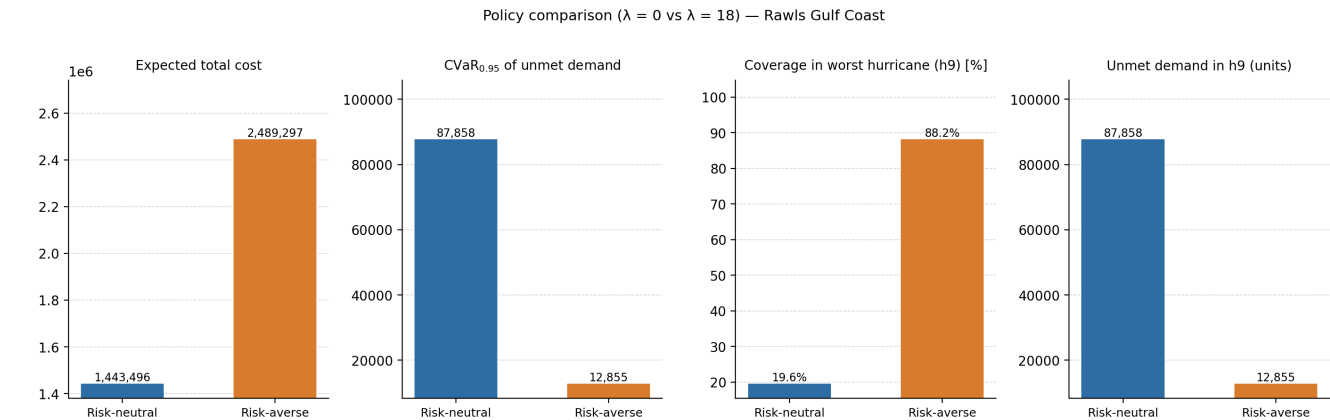


Figure 3: Four-panel comparison of the risk-neutral ($\lambda = 0$) and risk-averse ($\lambda = 18$) policies on expected total cost, CVaR_{0.95} of unmet demand, coverage in the worst hurricane (h9), and unmet demand in h9.

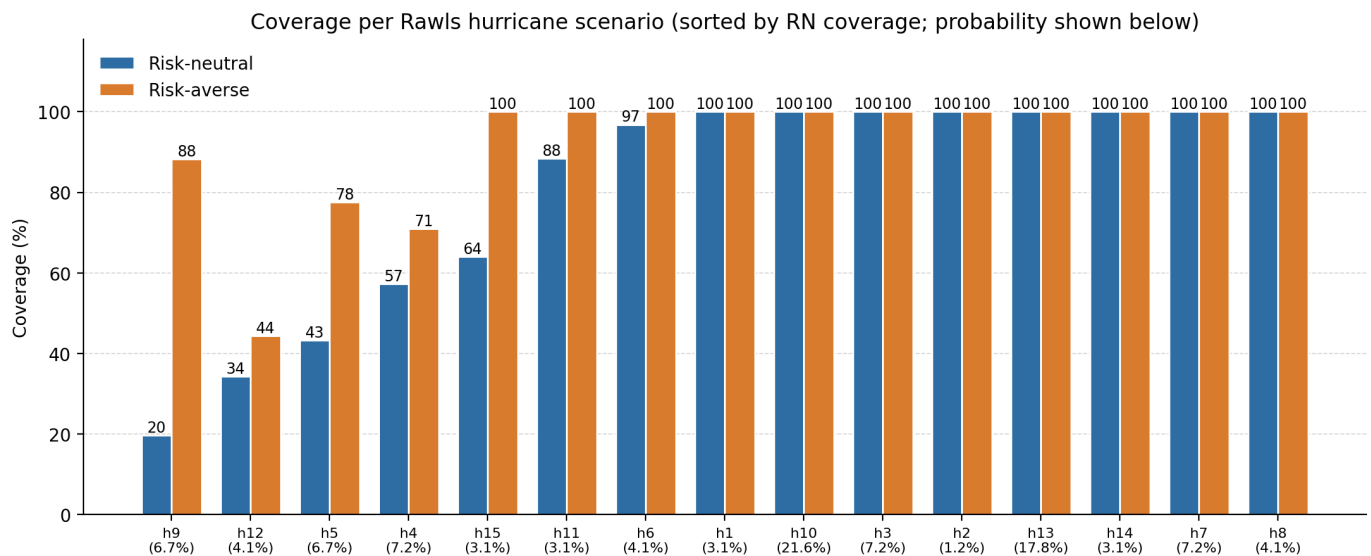


Figure 4: Coverage per hurricane, sorted by risk-neutral coverage. Scenarios are labelled with their renormalised probability. The risk-averse policy closes essentially every tail gap.

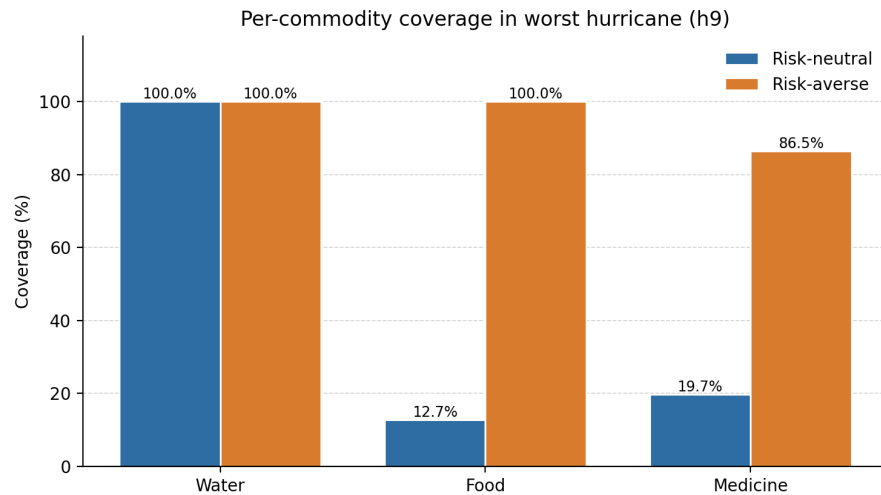


Figure 5: Per-commodity coverage in the worst hurricane (h9). The risk-averse policy in particular closes the gap on food and medicine, which drive the shortage volume.

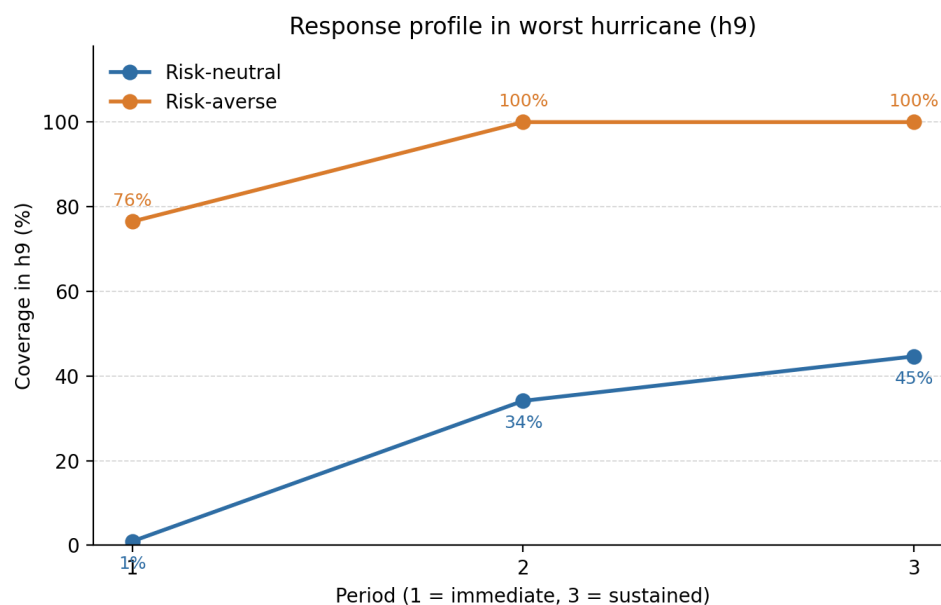


Figure 6: Period-specific coverage in the worst hurricane (h9). The risk-averse policy delivers coverage almost immediately in period 1, whereas the risk-neutral policy relies on slower catch-up in later periods.

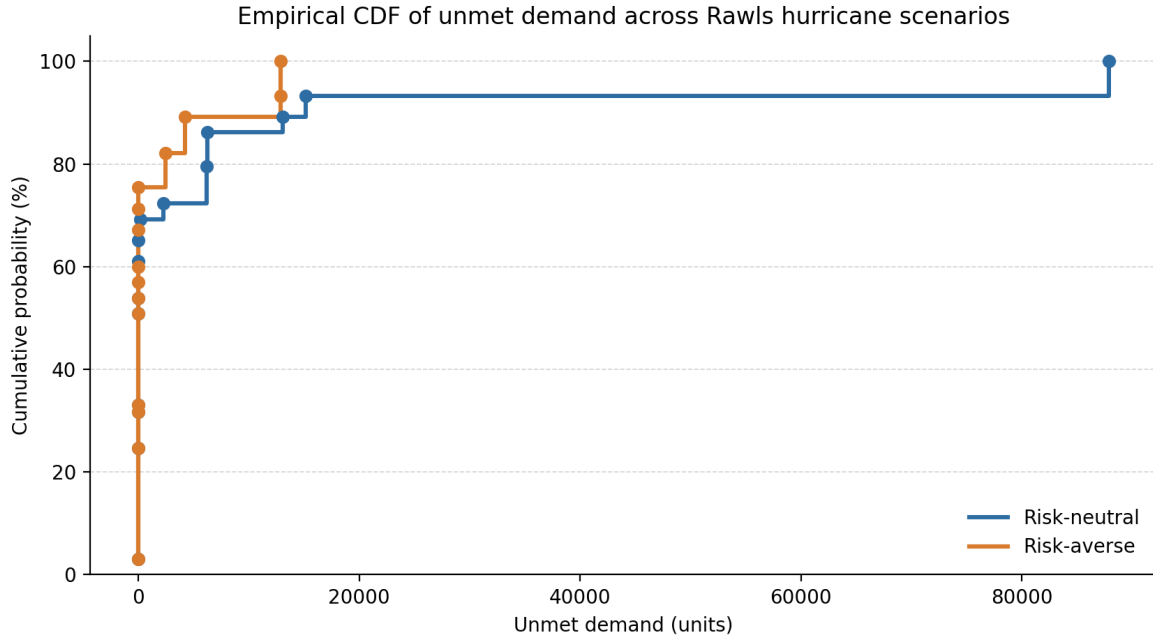


Figure 7: Empirical CDF of unmet demand across the fifteen Rawls hurricane scenarios. The risk-averse curve rises faster and saturates at a much smaller shortage level, which is exactly the CVaR improvement reported in Table 7.

The risk-averse policy opens four hubs (including the coastal candidate at node 22 that the risk-neutral policy leaves closed) and pre-positions substantially more stock. Expected cost rises by 72%, but $\text{CVaR}_{0.95}$ of unmet demand drops from 87,858 to 12,855 units, an 85% reduction. Coverage in the worst hurricane (h9, Category 5, striking nodes 13 and 29 with 95,000 units of medicine demand) jumps from 19.6% to 88.2%, and the period-1 share of that coverage rises from below 1% to 76.5%. Because the tail of the shortage distribution is dominated by h9 and h12 (the two hurricanes with the largest medicine and water demand), the $\text{CVaR}_{0.95}$ figure coincides with h9’s total unmet demand at the chosen α . Figure 7 makes the same point visually. Under the risk-neutral policy the CDF stays flat until large shortage levels appear, while under the risk-averse policy almost all probability mass sits at low unmet demand.

6.3 λ -Sensitivity

Figure 8 traces expected cost against $\text{CVaR}_{0.95}$ of unmet demand for $\lambda \in \{0, 5, 10, 15, 18, 22\}$. Higher λ shifts solutions toward lower tail shortages at the expense of higher expected cost, giving planners a tunable efficiency and resilience frontier. The frontier has two clear regimes. In the shallow region ($\lambda \leq 15$) marginal risk aversion trims a few thousand units of tail unmet demand for a modest cost bump. In the sharper region ($15 < \lambda \leq 22$) the model opens an extra hub and virtually eliminates the tail.

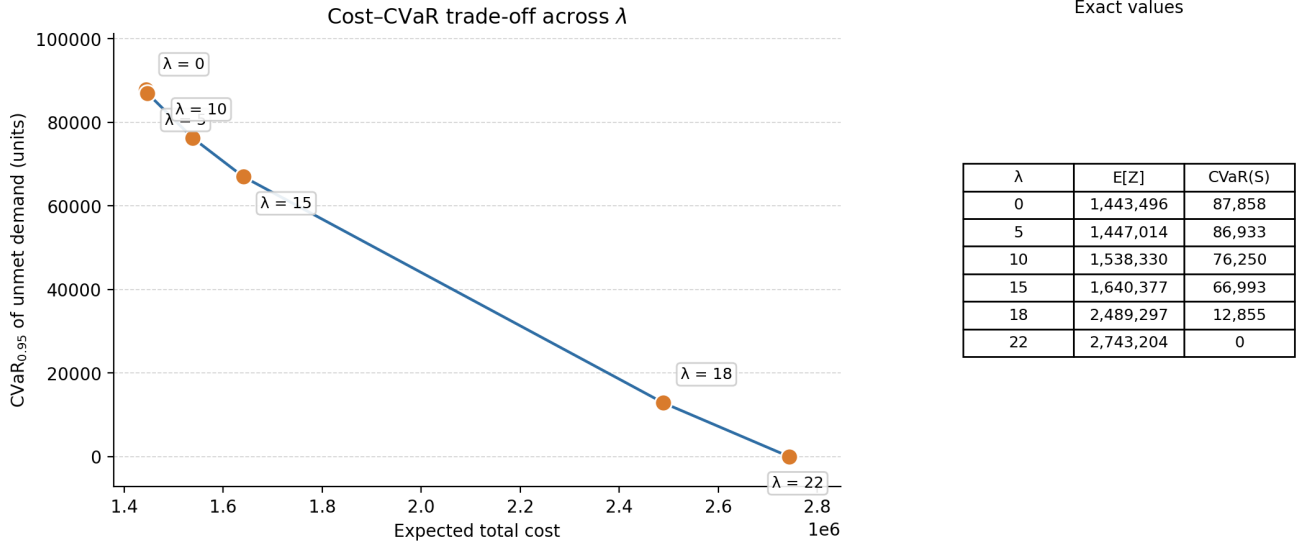


Figure 8: Expected cost against $CVaR_{0.95}$ of unmet demand across $\lambda \in \{0, 5, 10, 15, 18, 22\}$. The table on the right shows the exact CSV values.

7 Managerial Insights

The Rawls case study yields four observations that translate directly into planning guidance.

First, tail coverage can be bought for a bounded cost premium. Moving from $\lambda = 0$ to $\lambda = 18$ raises expected cost by 72% but reduces $CVaR_{0.95}$ of unmet demand by 85%. Planners who face a societal or legal obligation to guarantee minimum service in a worst-case hurricane can quantify that obligation as a specific λ on the Pareto curve rather than treating it as an open-ended commitment.

Second, tail improvements come from opening more hubs, not from bigger hubs alone. The risk-averse policy adds two additional candidate hubs (including one coastal site) rather than upsizing the two the risk-neutral policy already opens. Network breadth beats single-site depth once a hurricane can isolate a region.

Third, per-commodity reporting is essential. Aggregate coverage in the worst hurricane conceals the fact that water is already fully covered by the risk-neutral policy, whereas food and medicine coverage collapse. A manager who monitors only aggregate coverage would under-appreciate the vulnerability that CVaR exposes.

Fourth, response-time gains are largest in period one. The risk-averse policy delivers 76.5% of demand in the first period of the worst hurricane, versus under 1% for the risk-neutral policy. In humanitarian settings where mortality rises with delay, this early-period gain matters more than the same total volume delivered later.

8 Conclusion and Outlook

This paper formulated a two-stage stochastic mixed-integer programme for humanitarian pre-positioning and distribution and extended it with a mean-CVaR objective on total unmet demand, multi-mode transport (truck and air) with mode-specific delivery delays, and a multi-period response horizon. The extension combines the pre-positioning framework of Rawls and Turnquist [1], the multi-period logistics perspective of Yi and Özdamar [2], and the risk-averse stochastic programming approach of Noyan [3] into a single model that is deployable on a laptop-scale Gurobi licence.

Applied to the Rawls Gulf Coast case study with its fifteen historical hurricanes, the model shows that shifting from a risk-neutral to a calibrated risk-averse policy substantially improves both tail service and early-period response, at a bounded and quantifiable increase in expected cost. Reporting the trade-off as a Pareto curve gives planners an actionable decision aid rather than a single opaque number.

Several extensions are natural. First, distributionally robust reformulations would replace the fixed scenario probabilities with an ambiguity set, at the cost of more elaborate algorithmic machinery. This was intentionally out of scope after instructor approval of the CVaR focus. Second, the model could be calibrated with additional real hurricane data and refined commodity economics. Third, dynamic re-routing along the lines of Yi and Özdamar [2] would let the second stage react to information that arrives during the operational horizon. Fourth, incorporating equity constraints that bound the worst-off region's coverage would sharpen the humanitarian focus beyond aggregate tail metrics.

A Notation Legend

Table 8 collects the symbols used in the mathematical model so that a first-time reader can look them up in one place. Full definitions appear in Section 4.

Table 8: Compact notation legend for the risk-averse humanitarian logistics MILP.

Symbol	Meaning
N, I, D, A	Nodes, candidate hubs, demand nodes, directed arcs.
K, T, S, L, M	Commodities, periods, scenarios, facility sizes, transport modes.
$p_s, d_{ikts}, \gamma_{iks}$	Scenario probability, demand, surviving-stock fraction.
U_{ijmst}, F_{il}, M_l	Mode capacity, fixed opening cost, facility storage capacity.
$b_k, q_k, c_{ijkmts}, e_k$	Storage need, acquisition cost, unit transport cost, capacity use.
h_k, ρ_k, τ_m	Holding cost, shortage penalty, delivery delay by mode.
α, λ	CVaR level (0.95) and risk-aversion weight.
$y_{il}, r_{ik}, x_{ijkmts}$	Open hub, pre-position stock, multi-mode flow.
I_{ikts}, w_{ikts}	Inventory and unmet demand.
Z_s, S_s, η, ξ_s	Scenario cost, scenario shortage, VaR threshold, CVaR excess.

References

- [1] Rawls, C. G., and Turnquist, M. A. (2010). Pre-positioning of emergency supplies for disaster response. *Transportation Research Part B: Methodological*, 44(4), 521–534.
- [2] Yi, W., and Özdamar, L. (2007). A dynamic logistics coordination model for evacuation and support in disaster response activities. *European Journal of Operational Research*, 179(3), 1177–1193.
- [3] Noyan, N. (2012). Risk-averse two-stage stochastic programming with an application to disaster management. *Computers & Operations Research*, 39(3), 541–559.
- [4] Rockafellar, R. T., and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2, 21–41.